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RViT-FusionNet: A local cross-attention feature fusion-based hybrid framework for brain tumor classification

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Abstract
Accurate brain tumor classification via MRI is essential for diagnosis and treatment. This study introduces RViT-FusionNet, a hybrid deep learning model that integrates convolutional and transformer architectures for enhanced tumor detection. The model utilizes ResNet-50 to capture textural details and a Vision Transformer for extracting global context. A Local Cross-Attention (LCA) module is proposed to align and merge these features, allowing the network to model local structures and long-range dependencies concurrently. To enhance generalization across varied imaging conditions and tumor types, a domain discriminator is included to discern spatial and domain-specific patterns, fostering the learning of domain-invariant representations. The approach is validated on four public MRI datasets, yielding classification accuracies of 99.08% ± 0.16%, 99.56% ± 0.17%, 96.20% ± 0.25% and 94.76% ± 0.35% for glioma, meningioma, pituitary tumors, and healthy cases, respectively. For interpretability, Grad-CAM is utilized to create saliency maps highlighting tumor regions, confirming the model's focus on clinically relevant areas. These findings demonstrate that RViT-FusionNet achieves exceptional performance while maintaining high interpretability, positioning it as an effective tool for computer-assisted brain tumor diagnosis. This framework is posited as a robust and scalable solution for multi-class brain tumor classification in clinical practice. The code is available at https://github.com/naimaislam-ice/RViT-FusionNet-LCA-based-brain-tumor-classification.

Keywords Brain tumor classification · MRI · RViT-FusionNet · Local Cross-Attention · Grad-CAM

1 Introduction

Brain tumors are anomalous tissue growths that develop inside the brain or surrounding tissues. They can be either benign (non-cancerous) or malignant (cancerous), and they can start in the brain or spread to other areas of the body. Gliomas, meningiomas, and pituitary adenomas are common types of brain tumors. Meningiomas are often benign and develop from the protective membranes of the brain, whereas gliomas, which are frequently aggressive, originate from glial cells. The pituitary gland is the site of pituitary adenomas, which can throw off the hormonal balance. Depending on the type and location of the tumor, symptoms can include headaches, seizures, and problems with eyesight or hormones. As a result, brain tumors are a severe condition that can impact people of all ages. It is regarded as one of the deadliest illnesses in the world because of its extensive effects [1].

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This cancer is the third most common cancer in teenagers and young adults and is particularly common in older persons [2].

Magnetic Resonance Imaging (MRI) scans are frequently used for the diagnosis and detection of brain tumors [3]. MRI creates fine-grained images of the brain by combining radio waves and powerful magnetic fields. Generally speaking, tumors show up as areas of abnormally high signal intensity relative to the surrounding healthy brain tissue. These scans aid in determining the size, location, and extent of the brain tumor. Furthermore, tumor visibility is improved by contrast-enhanced MRI, in which a contrast agent (a unique dye) is administered intravenously either before or during the scan. Advanced MRI methods like diffusion tensor imaging (DTI) and functional MRI (fMRI) offer more information on the tumor's properties and how it affects brain connectivity and function [4].

It is difficult to classify brain tumors effectively since it involves taking into account several factors, including the tumor's size, histological grade, cell origin, anatomical location, molecular profiles, genetic mutations, and other distinctive features. Computer-aided categorization techniques have been created to deal with this complexity. Medical professionals can make effective treatment plans, obtain a better understanding of tumor kinds, and provide more accurate prognostic assessments by utilizing these AI-driven tools [5]. The detection and categorization of brain tumors has made extensive use of AI and computer vision techniques in recent decades. Through the identification of pathological traits and the support of physicians in making well-informed judgments, these automated methods seek to increase diagnostic accuracy and lower total healthcare costs [6]. Advanced MRI analysis techniques, for example, use metabolic signals from proton MR spectroscopy or picture texturing to assist in differentiating between different types of tumors [7]. By automatically learning intricate imaging patterns, deep learning, particularly multi-layer artificial neural networks, has transformed the classification of brain tumors [5].

Convolutional neural networks, or CNNs, are particularly popular for medical image processing and are now a common method for classifying brain tumors [8–12]. Several convolutional layers are used in a conventional CNN architecture to extract local characteristics from input images. These features are then synthesized by fully connected layers to generate a final classification. Many teams use MRI data to refine ImageNet pre-trained CNNs. ResNet-50, Inception V3, and VGG-16 are a few examples of feature extractors used in their brain tumor system. Inception's multi-path modules capture multi-scale patterns, VGG's simple repetitive convolutions offer fine detail, and ResNet's residual blocks enable very deep networks. These deep nets have complementary strengths. Other designs, such as DenseNet, Xception, and MobileNet, have also been used to analyze MRI tumor data [13]. In medical imaging, Vision Transformers (ViTs) have recently become a viable substitute for CNNs [14]. ViTs divide an image into patches and apply self-attention to each patch, in contrast to CNNs. Long-range, global relationships in the image are captured by this attention process. Dependencies across distant regions, instead of only local features, are one example. In reality, a ViT learns visual features by first linearly projecting each image patch to tokens and then using a transformer encoder block. This global context can assist the model in taking into account minor patterns that extend throughout the tumor and surrounding tissue in the brain MRI domain [15]. Designing a hybrid model is also inspired by vision transformers. To achieve the best of both worlds, some models combine elements of Transformer and CNN. For example, Swin-based architectures employ shifted-window attention modules with residual MLPs, whereas ARiViT mixes residual transformer blocks with convolutional encoding [16, 17]. These hybrid networks can record global attention patterns and local CNN-style features at the same time. Preliminary findings indicate that these models enhance classification resilience and accuracy.

To achieve this, we present RViT-FusionNet, a hybrid architecture that combines a Vision Transformer (ViT) branch for capturing global contextual information with a ResNet-50 backbone for extracting fine-grained local texture features such as tumor boundaries, edges, and subtle tissue variations. The integration of these two complementary feature representations is facilitated by a Local Cross-Attention (LCA) module, which aligns multi-scale transformer tokens with convolutional feature maps. This module enables each local patch descriptor to selectively attend to relevant global contexts, thereby refining both local and global features for improved

classification performance. To ensure robustness across varying MRI acquisition protocols, this architecture incorporates an adversarial domain discriminator, which promotes domain-invariant feature learning while preserving discriminative power. Furthermore, this study employs Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize and localize tumor regions, enhancing model interpretability by highlighting the spatial areas in the MRI scan that contribute most significantly to the classification decision. An overview of this proposed method is illustrated in Fig. 1. This work contributes to this field of brain tumor classification in several ways:

- The proposed method, RViT-FusionNet, combines the strengths of ResNet-50 and Vision Transformer for brain tumor classification using MRI images by introducing a Local Cross-Attention (LCA) module that fuses local texture features from ResNet-50 with global context features from ViT, thereby enhancing spatial understanding and feature alignment.
- This study presents a patch-based domain discriminator that utilizes adversarial learning with binary cross-entropy loss to derive domain-invariant spatial features from ResNet-derived maps, improving the robustness and generalizability of RViT-FusionNet across various MRI scanners.
- Several brain MRI datasets are used to evaluate the proposed strategy, showing its efficacy while keeping the test data entirely hidden from the model.
- This research uses Grad-CAM to localize tumor regions in MRI scans that influenced the classification, thus enhancing trust and transparency in the model's decision-making process.

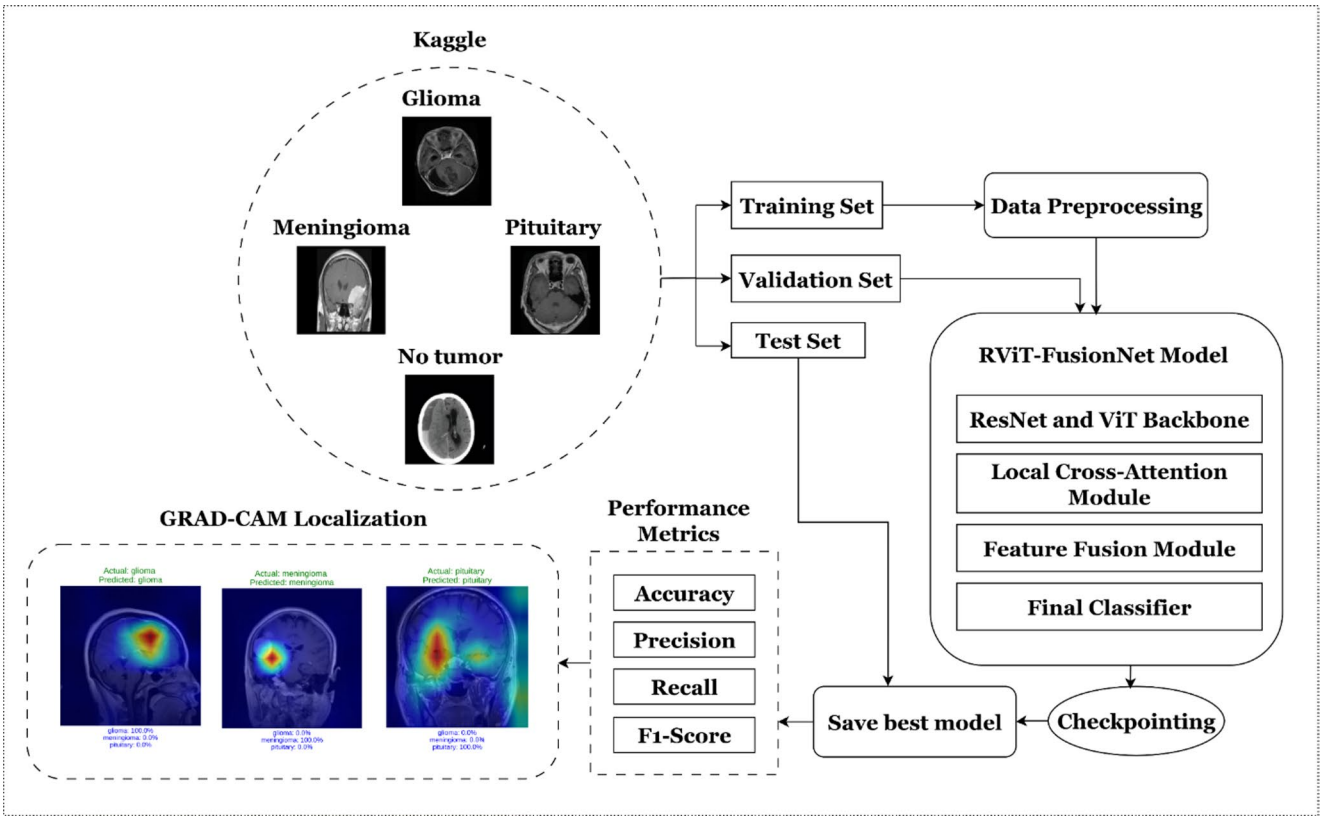


Fig. 1 An overview of the proposed RViT-FusionNet method for brain tumor classification, illustrating dataset preparation, model architecture, performance evaluation, and Grad-CAM-based localization

2 Literature review

2.1 Convolutional neural network-based models

CNNs are widely used in image analysis for their ability to learn hierarchical features. In medical imaging tasks like brain tumor detection, fine-tuning pre-trained CNNs is common, where final layers are adapted to the target task, enabling effective learning from limited medical data [18]. Ishfaq et al. classified brain MRI data into ten tumor types using a custom CNN along with Inception-v4 and EfficientNet-B4, showing strong performance across multiple datasets [19]. Ishaq et al. developed an EfficientNet-based model with Grad-CAM for explainability, classifying four tumor types while reducing complexity [20]. Nazir et al. designed a CNN enhanced with SHAP, LIME, and Grad-CAM, achieving near-perfect accuracy on the BR35H dataset and improving model interpretability [21]. Iftikhar et al. created a simpler explainable CNN using Grad-CAM, SHAP, and LIME, balancing accuracy with interpretability [22]. Srinivasan et al. integrated a channel-wise attention mechanism into ResNet-101 to boost the classification of four tumor classes [23]. Reyes and Sánchez compared various CNNs, showing that lightweight models like MobileNet and EfficientNet can offer competitive accuracy with lower computational cost, especially with transfer learning [24]. Nikhileswar et al. proposed an ensemble of ResNet, VGG, and MobileNet, achieving high multi-class tumor detection accuracy [25].

2.2 Transformer-based models

Vision Transformers (ViTs) model long-range dependencies through self-attention and differ from CNNs in feature extraction. Wang et al. proposed RanMerFormer, a ViT-based pipeline with token merging and a randomized functional-link layer that achieved state-of-the-art classification on public MRI datasets [26]. Nalentzi et al. fine-tuned ViT and BEiT models on semi-automatically segmented tumor patches for accurate glioma and meningioma classification [27]. Simon and Briassouli trained pure ViT models from scratch on brain tumor MRIs and found their performance to be highly dependent on dataset size [28]. Asiri et al. developed FT-ViT, which uses position embeddings and self-attention to detect global tumor patterns in MRI data [29]. Tummala et al. fine-tuned multiple ViT variants and combined them into an ensemble to improve robustness in multi-class tumor classification [30]. Kumar Reddy et al. compared four fine-tuned ViT variants and confirmed their capability in distinguishing glioma, meningioma, pituitary, and normal classes [31]. Hussein Ali introduced ViT-BT, a transfer learning approach that achieved superior classification on the BraTS dataset using multi-sequence MRI tumor data [32]. Hossain et al. introduced the IVX16 model, achieving accurate and interpretable brain tumor classification [33]. Pacal et al. introduced NeXtBrain, a hybrid CNN-Transformer model that combines self-attention for global context with convolutional modules for fine-grained local texture extraction in 3D MRI scans [34]. Panigrahi et al. proposed DenseTransformer, which integrates transformer-based self-attention layers with a DenseNet201 backbone to capture both long-range dependencies and local spatial features [35]. Goceri developed a CNN-Transformer hybrid that uses attention modules between CNN layers to focus on tumor-related regions while maintaining computational efficiency [36]. Jin et al. presented EMCAH-Net, a multi-scale CNN-Transformer network that captures cross-scale relationships to improve abnormal tissue localization [37]. Cui et al. designed ResMT, which uses 3D CNNs for local morphology extraction and stacked self-attention transformers for modeling global spatial patterns in glioma grading [38]. Tabatabaei et al. created a two-branch CNN-Transformer model with a cross-fusion procedure that combines local CNN features and global Transformer features for multi-class tumor classification [39].

Brain tumor classification from MRI is essential for accurate diagnosis and treatment planning. A summary of related works is presented in Table 1. Recent advances in deep learning including Convolutional Neural Networks (CNNs), Residual Networks (ResNet), and Vision Transformers (ViT) have improved classification by learning hierarchical and contextual image representations; however, each approach has limitations: CNNs capture local features well but struggle with long-range dependencies, while Transformer models global context effectively but

Table 1 Summary of the Existing Approaches for Brain Tumor Classification from MRI images

Type	References	Method	Dataset (No. Images)	Classes	Limitations
Convolutional Neural Network-based Models	Ishfaq et al. [19]	Custom convolutional neural network (CNN) model	Several publicly available datasets in Kaggle	10	Lack of generalization, bias in the dataset, and need for clinical validation.
	Ishaq et al. [20]	Improved version of EfficinetNet model	Brain Tumor MRI Dataset (BTMD), and NeuroMRI (7023, 3264)	4	Failure to incorporate clinical or patient history data.
	Nazir et al. [21]	Customized CNN model	BR35H (3060)	2	Only binary classification is possible; multiple tumor types cannot be distinguished.
	Iftikhar et al. [22]	XAI-based CNN model	BTMD, and NeuroMRI (7023, 3264)	4	Lower accuracy on unseen data.
	Srinivasan et al. [23]	ResNet101-CWAM	BTMD (7023)	4	Larger model size due to CWAM modules.
Transformer-based models	Wang et al. [26]	RanMerFormer	BTMD, figshare Brain Tumor Dataset (FBTD) (7023, 3064)	4, 3	High computational cost.
	Nalentzi et al. [27]	ViT and BEiT	Public (1650)	2	Confined to just two types of tumors.
	Simon and Briassouli [28]	ViT model	NeuroMRI (3264)	4, 2	Overfitting results from inadequate regularization and augmentation.
	Asiri et al. [29]	FT-ViT	BTMD (7023)	4	Lack of interpretability of the model's predictions and the outcome.
	Tummala et al. [30]	Ensemble ViT models	FBTD (3064)	3	Model's generalizability not shown on an external dataset.
	k. Hussein Ali [32]	ViT-BT model	BraTS-2023	1	Limited training data.
	Hossain et al. [33]	IVX16 model	Proprietary dataset (3264)	4	High complexity
	Pacal et al. [34]	NeXtBrain model	BTMD, FBTD (7023, 3064)	3, 4	Limited generalizability, Dataset dependency
	Panigrahi et al. [35]	DenseTransformer model	BR35H (3060)	2	Ignored subtypes, Limited applicability.
	E. Goceri [36]	CNN-Transformer	BraTS-2020	1	Lack of attention to subgroups and limited relevance.
	Cui et al. [38]	ResMT model	BraTs19 dataset	1	Insufficient consideration for subgroups and restricted applicability.
	Tabatabaei et al. [39]	HMAC-Net	BraTS2021, AANLIB, ISIC2018 (1251, 432 and 2112)	----	Low accuracy, Limited reliability

typically need large datasets. To address these complementary weaknesses, we propose RViT-FusionNet, a hybrid architecture that fuses ResNet and ViT components to jointly capture local and global features from MRI scans, improving classification accuracy and robustness.

3 Materials and methods

3.1 Experimental setup

We utilized a T4 × 2 GPU in the Kaggle environment to build a high-performance system for brain tumor classification. An Intel Core-i5 processor with 8 GB of RAM and a 4 GB local GPU powers this setup on a Dell Inspiron.

The PyTorch framework was used for model development and training because of its versatility and efficiency in handling deep learning tasks.

3.2 Dataset

Four publicly accessible brain MRI datasets, designated as DS1, DS2, DS3 and DS4, were utilized to construct and critically assess the proposed RViT-FusionNet model.

- A collection of three datasets—Figshare, SARTAJ, and Br35H—DS1 is accessible on the Kaggle website [51]. There are four different kinds of MRIs in DS1: pituitary, glioma, meningioma, and no tumor. DS1 has 7023 images in total, of which 1311 are being set aside for testing and 5712 for training. Sample images of this dataset are shown in Fig. 2(a).

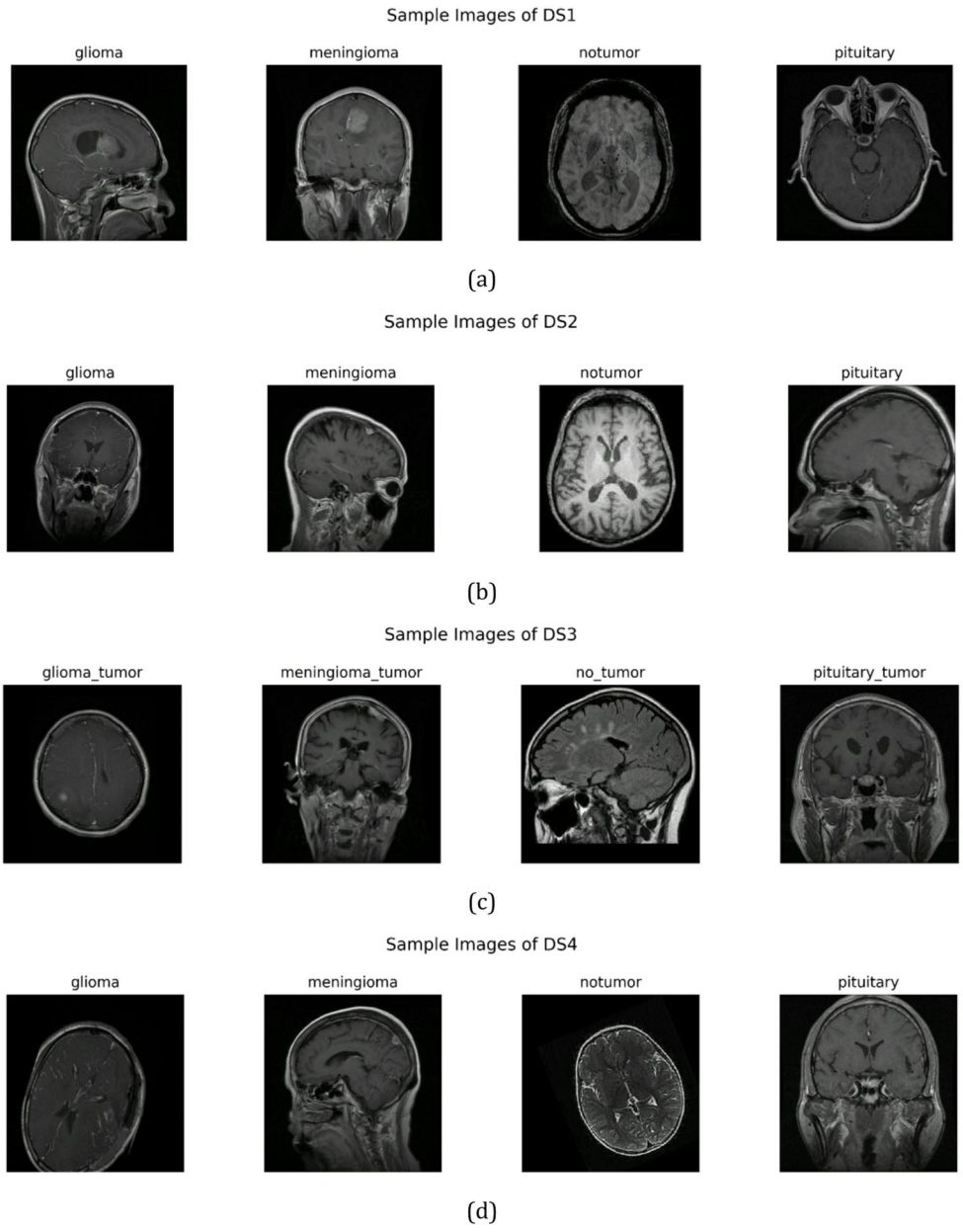


Fig. 2 Sample images from different datasets. (**a**) DS1, (**b**) DS2, (**c**) DS3, and (**d**) DS4

- The dataset DS2 can be accessed via the Kaggle platform [52]. DS2 is another version of DS1 that includes additional images, and it comprises 7153 images. However, there is some overlap between the two datasets. Like DS1, DS2 contains four categories of MRI images, namely glioma, meningioma, pituitary, and no tumor. A selection of images from DS2 is displayed in Fig. 2(b).
- The DS3 dataset is accessible on Kaggle [53]. There are four distinct types of MRIs: pituitary, meningioma, glioma, and no tumor. There are 394 images in DS3 for testing and 2870 images for training. However, we combine the two sections and use the entire set of pictures to evaluate the proposed model. Several images of this dataset are shown in Fig. 2(c).
- The dataset DS4 is accessible on Kaggle [54]. There are four different categories of MRIs: glioma, meningioma, pituitary, and no tumor. There are 32,080 images in DS4 for training and 8020 images for testing. In total, this dataset comprises 40,100 images. So, we combine the two sections and use the entire set of images to evaluate RViT-FusionNet. Figure 2(d) displays a sample of images from this dataset.

We trained RViT-FusionNet on the training portion of DS1 and selected the best model based on validation results obtained during training. The best model was then saved for later evaluation on the DS1 test set as well as on the other three datasets: DS2, DS3, and DS4.

3.2.1 Data augmentation

The model was successfully trained for precise brain tumor classification in the current study using a variety of data preparation techniques. These preparation steps improve the model's generalization to unknown medical imaging data, which is essential for creating trustworthy diagnostic systems. Figure 3 illustrates various data augmentation techniques applied to a brain MRI scan. Through the introduction of controlled fluctuations in the input images, the preprocessing pipeline decreased overfitting during training, enhanced resilience, and guaranteed consistency. The following actions were part of the transformation techniques:

- Training Transformations: To add variability and improve the robustness of the model, the following augmentations were used during training:
  - Random Resized Crop: Images were resized to 224×224 pixels and cropped at random using a scale range of (0.8, 1.0) to replicate changes in object position and size.
  - Random Horizontal Flip: Images were 50% likely to be flipped horizontally.
  - Random Vertical Flip: To enhance spatial invariance, images were flipped vertically with a 50% chance.
  - Random Rotation: The rotational resilience of the images was increased by rotating them by up to 15 degrees.

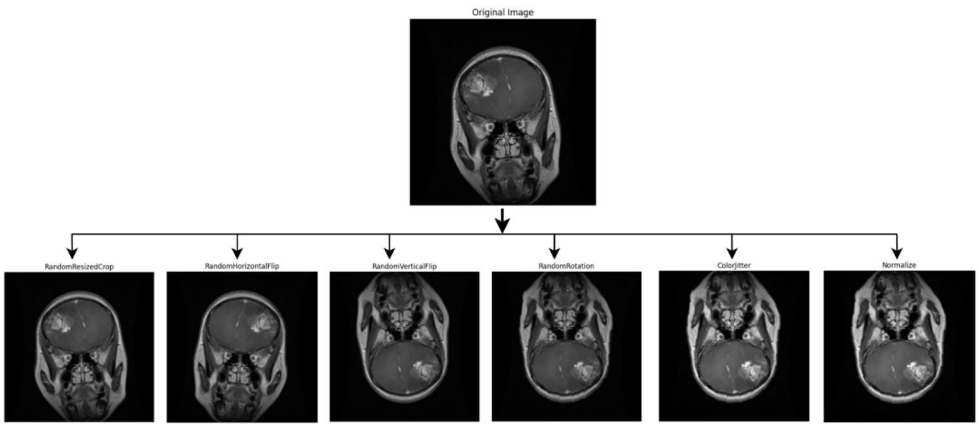


Fig. 3 Data Augmentation Techniques Applied to a Brain MRI Image

  - Color Jitter: The brightness and contrast were randomly adjusted by (±30%) to replicate a range of lighting conditions.
  - ToTensor: PyTorch tensors were used to convert image data, scaling pixel values from [0, 255] to [0, 1].

- Normalization: Based on ImageNet statistics and standardized using mean values [0.485, 0.456, 0.406] and standard deviations [0.229, 0.224, 0.225].

- Validation and Test Transformations: Only deterministic alterations were applied to the test and validation sets to guarantee consistent evaluation:

  - Resize: Images were resized to a fixed size of 224 × 224 pixels.
  - ToTensor: Images were converted to PyTorch tensors.
  - Normalization: For uniform input scaling, the same mean and standard deviation values were used as in training.

For accurate performance evaluation, these preprocessing pipelines made sure the model was tested on standardized data after being trained on a variety of inputs.

3.3 Multiscale feature fusion structure

Effectively integrating local and global feature learning is crucial for accurate diagnosis when classifying brain tumors from MRI data. Convolutional layers capture local features, which emphasize fine-grained details like tumor margins, texture variations, and subtle tissue irregularities. Global features, which are obtained through spatial pooling or attention mechanisms, encode the broader anatomical context, which includes the tumor's position with surrounding brain structures. These two feature streams are combined in a specific module in the suggested design, which aligns and combines comprehensive image representations with spatial cues to create a single feature vector. RViT-FusionNet may therefore use both detailed local information and thorough contextual understanding due to this enriched representation, which is then supplied to a classification head. By utilizing this hybrid architecture, the system improves tumor classification accuracy and robustness by becoming more sensitive to subtle pathological alterations while maintaining a sense of the whole scan context. The multiscale feature fusion architecture is shown in Fig. 4.

3.4 Proposed method

This study presents RViT-FusionNet, a novel hybrid architecture that has been painstakingly created to use convolutional neural networks and transformer-based models to their complementary strengths in order to classify brain tumors from MRI data. The architecture uses a ResNet-50 backbone to record tumor border details, fine-grained tissue changes, and edges, among other rich local texture characteristics. Concurrently, a Vision Transformer (ViT) branch is employed to encode extensive contextual information and long-range dependencies throughout the scan, guaranteeing a thorough comprehension of the input. In order to bridge these two

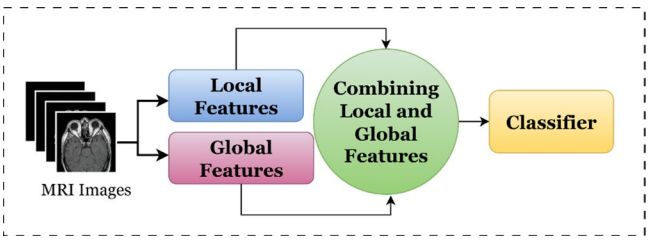


Fig. 4 Block diagram of the proposed multiscale feature fusion architecture

representational paradigms, this work includes a Local Cross-Attention module that aligns transformer tokens at different spatial resolutions with convolutional feature maps. The attention weights that are calculated by this module allow each local patch descriptor to preferentially focus on pertinent global contexts. By doing this, it refines global representations with localized information and enhances local descriptors with larger anatomical signals. Moreover, the proposed model incorporates an adversarially trained domain discriminator to provide resilience against changes in MRI acquisition techniques and scanner properties. By promoting the learning of domain-invariant feature embeddings, this component maintains the discriminative capability necessary for precise tumor classification while rendering them unrecognizable between source and target domains. In brain tumor classification tasks, RViT-FusionNet's harmonious integration of local and global variables ensures both precision and generalizability thanks to its well-thought-out design. To visually interpret the decision-making process and highlight the tumor regions, we utilize Grad-CAM, which helps localize critical areas contributing to the classification. The RViT-FusionNet system architecture is depicted in Fig. 5.

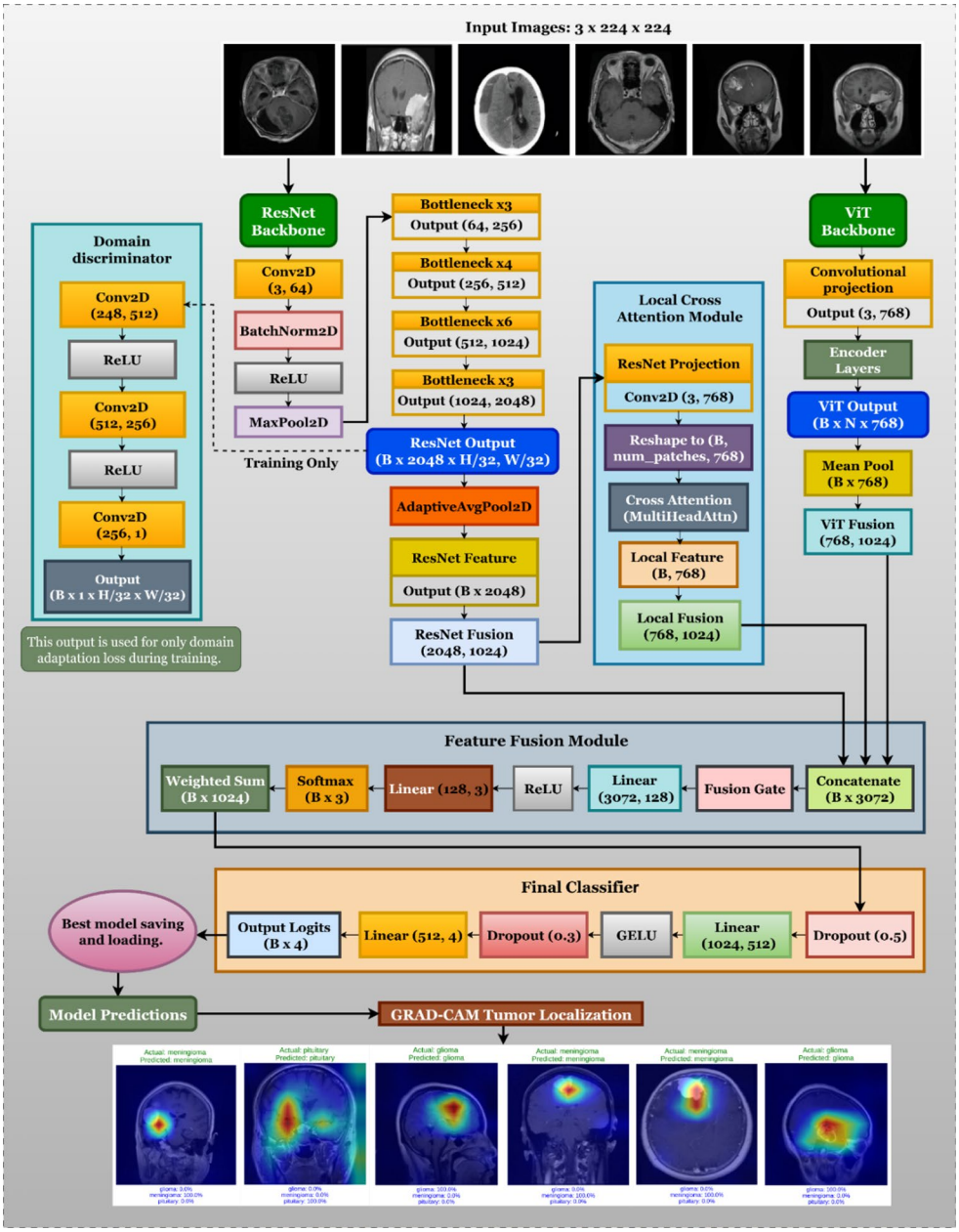


Fig. 5 Proposed system architecture for brain tumor classification, where the Grad-CAM shows the location of each tumor (best seen in color)

3.4.1 ResNet-50 backbone

The ResNet-50 backbone in the proposed architecture reduces the spatial resolution by processing each input MRI image through a sequence of bottleneck and convolutional layers, including a convolution layer, batch normalization, and max pooling, and then four successive bottleneck blocks with output channel sizes of 256, 512, 1024, and 2048, respectively. The network can extract rich local texture features, including edges, fine-grained tissue variations, and tumor boundaries, thanks to the hierarchical structure of these layers, and the result is a high-dimensional feature map of shape $B \times 2048 \times \frac{H}{32} \times \frac{W}{32}$, where B is the batch size, and H and W represent the input height and width. An adaptive average pooling layer shrinks the spatial dimensions to 1×1 to produce a pooled feature vector with the shape $\times 2048$, which provides a compact global representation for tasks that come after. The pooled feature vector can be expressed mathematically as follows:

$$f_{resnet} = AdaptiveAvgPool\left(F_{resnet}\right) \in \mathbb{R}^{B \times 2048} \tag{1}$$

where, $F_{resnet} \in \mathbb{R}^{B \times 2048}$ is the output feature map from the final ResNet layer.

3.4.2 ViT backbone

The first step for the ViT backbone is to split the input image into 16×16 non-overlapping patches. To create a sequence of $N = \frac{H}{16} \times \frac{W}{16}$ patch embeddings, each patch is linearly embedded into a 768-dimensional vector using a trainable projection layer. Modeling long-range dependencies and contextual interactions across the scan, these embeddings are processed by numerous transformer encoder layers. The ViT branch produces a tensor with the shape $\times \times 768$, where is the total number of patches. The patch embeddings are mean-pooled along the sequence dimension to provide a final feature vector with the shape $\times 768$, which is the result of a comprehensive global feature representation. In formal terms, the global ViT feature vector is calculated as follows:

$$f_{vit} = \frac{1}{N} \sum_{i=1}^{N} P_i \in \mathbb{R}^{B \times 768} \tag{2}$$

where, $P_i \in \mathbb{R}^{B \times 768}$ indicates the i-th patch's embedding. The ViT branch is guaranteed to capture thorough contextual information due to this procedure.

3.4.3 Local cross attention module

The Local Cross-Attention (LCA) module is a crucial fusion mechanism that combines the global contextual representations generated by a Vision Transformer (ViT) with the intricate local information collected by a ResNet-50 while rigorously maintaining the ViT's internal patch embeddings. The ResNet outputs a tensor $R \epsilon \mathbb{R}^{B \times 2048 \times H \times W}$, where B is the batch size, 2048 channels represent local feature richness and $H \times W$ the spatial dimensions. This tensor is projected into the ViT embedding space of 768 dimensions by applying a learnable 1×1 convolutional layer, yielding $R' \epsilon \mathbb{R}^{B \times 768 \times H \times W}$. The resulting projection is then reshaped into a sequence of query vectors, $Q = reshape\left(R'\right) \epsilon \mathbb{R}^{B \times (H \times W) \times 768}$ that represents all spatial locations. The ViT patch embeddings serve as the fixed keys and values with shape:

$$K = V \epsilon \mathbb{R}^{B \times L \times 768} \tag{3}$$

Where, L is the number of patches. These local ResNet queries then engage with the global ViT features, which serve as both keys K and values V, through scaled dot-product attention:

$$Attention\left(Q,\ K,\ V\right) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) V \epsilon \mathbb{R}^{B \times (H \times W) \times 768} \tag{4}$$

where, $d_k = 768;$ denoting the embedding dimension.

This attention mechanism is asymmetric in nature. So, ViT's keys and values remain constant, while the ResNet-derived queries adaptively attend to ViT's global feature context, thereby enriching local ResNet features with comprehensive semantic information. The attended features are subsequently aggregated across all spatial positions by mean pooling, which produces a single, rich vector that combines local texture details and global image-level information. The function is defined as:

$$F = \frac{1}{H \times W} \sum_{i=1}^{H \times W} \widehat{Q_i} \, \epsilon \, \mathbb{R}^{B \times 768} \tag{5}$$

The LCA module employs 8 attention heads, following the standard ViT-B/16 configuration to maintain compatibility with pre-trained weights. This established hyperparameter balances model capacity with computational efficiency without requiring extensive tuning. The module implements global attention where all ResNet-derived spatial queries interact with all ViT patch embeddings, eliminating explicit window restrictions. This design choice simplifies training by avoiding complex localized attention patterns while enabling comprehensive feature integration. The global attention mechanism allows each local feature to access relevant information from any part of the image through the transformer's contextualized representations. This approach facilitates maximal information exchange between convolutional and transformer pathways while leveraging proven architectural standards.

3.4.4 Domain discriminator

We use a Patch-based Auxiliary Discriminator linked to the ResNet50 backbone to enhance the resilience of spatial feature representations and model generalization across various MRI acquisition techniques while reducing scanner-specific bias. Since existing studies have shown that supervised domain adaptation frameworks can greatly improve performance in target domains with limited annotations [40], our design similarly aims to introduce robustness by encouraging the network to learn domain-invariant representations. We build this module as a stochastic regularizer, in contrast to traditional Domain-Adversarial Neural Networks (DANN), which depend on static source/target labels and Gradient Reversal Layers (GRL) to enforce feature invariance. The discriminator uses three convolutional layers to handle uncompressed feature maps $F_{res} \in \mathbb{R}^{B \times 2048 \times 7 \times 7}$ via a three-layer fully convolutional network: two 3×3 convolutional layers (reducing channels $2048 \rightarrow 512 \rightarrow 256$) followed by ReLU activations and a final 1×1 convolutional layer to project the output to a single channel. This results in a 7×7 grid of logits. Instead of using fixed domain identifiers, we specify the target labels as randomized binary values $\{0, 1\}$ created dynamically at each cycle. In order to prevent overfitting to the main classification task, the module introduces controlled noise into the backbone's updates using standard backpropagation without gradient reversal. This auxiliary Binary Cross-Entropy loss is integrated into the global objective with a weighting factor of $\lambda = 0.2$.

3.4.5 Feature fusion module

The suggested RViT-FusionNet includes a Feature Fusion Module that adaptively combines global and local representations to efficiently utilize the complementary advantages of convolutional and transformer-based features. Specifically, the global features extracted from the ResNet backbone ($r \in \mathbb{R}^{B \times 2048}$), the global ViT embeddings ($v \in \mathbb{R}^{B \times 768}$), and the locally enhanced features from the Local cross attention module ($l \in \mathbb{R}^{B \times 768}$) are each projected into a common fusion space of dimension 1024 using learnable linear layers:

$$r_f = W_r r, \quad v_f = W_v v, \quad l_f = W_l l \tag{6}$$

where,

$$W_r \in \mathbb{R}^{1024 \times 2048}$$

$$W_v \in \mathbb{R}^{1024 \times 768}$$

$$W_l \in \mathbb{R}^{1024 \times 768}$$

These projected features ($r_f, v_f, l_f \in \mathbb{R}^{B \times 1024}$) are then concatenated to form a combined feature vector ($f_{cat} \in \mathbb{R}^{B \times 3072}$) . A fusion gate, which is a two-layer multilayer perceptron (MLP) with ReLU activation and softmax normalization, is applied to f_{cat} in order to adaptively ascertain the contribution of each feature stream. This gate outputs a set of weights ($g \in \mathbb{R}^{B \times 3}$) for each sample, representing the importance of ResNet, ViT and local features respectively. The final fused representation is computed as a weighted sum:

$$f_{fused} = g_1 r_f + g_2 v_f + g_3 l_f \tag{7}$$

where, $_1$,$_2$,$_3$ are the softmax-normalized weights for each feature stream. This fused feature vector $f_{fused} \in \mathbb{R}^{B \times 1024}$ is then sent to the last classifier for the prediction of tumors. The Feature Fusion Module makes sure that the model can dynamically highlight the most informative representations for each input by learning to adaptively balance the influence of global and local features, improving classification performance and robustness.

3.4.6 GELU activation

The Gaussian Error Linear Unit (GELU) activation function is used after the first fully connected layer in the final classifier of the suggested RViT-FusionNet architecture to add a smooth, probabilistic nonlinearity. The mathematical definition of the GELU activation is:

$$GELU\,(x) = x.\varphi\ (x) \tag{8}$$

where, () is the standard normal distribution's cumulative distribution function (or CDF), which is provided by

$$\varphi\ (x) = \tfrac{1}{2}\ \left[1 + \mathrm{erf}\left(\tfrac{x}{\sqrt{2}}\right)\right] \tag{9}$$

A continuous gating mechanism is produced by GELU's capacity to evaluate inputs according to their likelihood of being positive under a Gaussian distribution. This smooth nonlinearity maintains tiny fluctuations in the feature space, which is essential for identifying fine-grained patterns in MRI brain tumor images, in contrast to the severe thresholding of ReLU. Additionally, backpropagation is improved by GELU, which lowers the possibility of dead neurons and improves training stability. The proposed hybrid model architecture design is a perfect fit for it due to its demonstrated efficacy in transformer topologies, which enhances classification performance and provides richer feature representations.

3.4.7 Parameters overview

Every significant part of the RViTFusionNet architecture, such as the classifier head, fusion layers, attention modules, and backbone streams, has its trainable parameters listed in detail in Table 2. As can be seen, the deep blocks of the ResNet and ViT backbones significantly increase the model's complexity as each extra bottleneck or transformer encoder adds millions of weights, while the Local Cross Attention and Feature Fusion submodules add a moderate amount of parameter overhead to efficiently integrate local and global representations. Despite its relatively small weight, the domain discriminator branch is expressive enough for adversarial adaptation. In order to ensure fine-grained tumor class separation, the densely coupled fusion gate and two-layer classifier head toward the network's terminus together consist of around 0.15 M parameters. The ability of RViT-FusionNet to

Table 2 Parameter overview of the proposed RViT-FusionNet

Module	Output Shape	Parameters
ResNet-50 Backbone	B × 2048 × 7 × 7	25.12 M
ViT Backbone (16 × 16)	B × 768	84.89 M
Local Cross-Attention Module	B × 1024	3.29 M
Domain Discriminator	B × 1 × 7 × 7	10.40 M
Feature Fusion Module	B × 1024	4.39 M
Final Classifier	B × 4 logits	0.15 M
Total Parameters: 128.45 M		

combine convolutional and transformer features for reliable brain tumor classification while striking a balance between representational power and computational tractability is demonstrated by its cumulative parameter count of 128,454,216.

3.5 Model training

The RViT-FusionNet architecture implemented for the classification of MRI brain tumors was trained with a batch size of 16 across both the training and validation processes. The model was trained using the training set of DS1, which comprises 5712 images categorized into glioma, meningioma, pituitary, and no tumor classes. The AdamW optimizer was meticulously configured with varying learning rates: 0.0010 designated for the classifier and fusion gate, 0.0005 allocated to the local cross-attention module, and 0.00001 assigned to the pre-trained ResNet and ViT backbone components. A cosine annealing scheduler was implemented to regulate the learning rate modifications throughout the training process, and model checkpoints were preserved based on validation accuracy to ensure the retention of the optimal model weights. In the case of RViT-FusionNet, the training iteration covered a predetermined number of epochs, 50. The model parameters were modified during each epoch in response to the loss obtained from the training data.

3.5.1 AdamW with differential learning rates

The AdamW optimizer is used to optimize the model parameters; it is ideal for transformer-based and deep convolutional models and separates weight decay from the gradient update. The various modules are allocated varying rates of learning:

- Fusion and Classifier Layers: Higher learning rate (η_{fusion}, $\eta_{classifier}$).
- Pretrained Backbones (ResNet-50, ViT): Lower learning rate ($\eta_{backbone}$).

The update rule for parameter θ at step t is:

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{\widehat{m_t}}{\sqrt{\widehat{v_t}} + \epsilon} - \lambda \cdot \theta_t \tag{10}$$

where, $\widehat{m_t}$ and $\widehat{v_t}$ are bias-corrected first and second moment estimates, η is the learning rate (varies per module), and λ is the weight decay coefficient.

3.5.2 Learning rate scheduling: Cosine annealing

A cosine annealing scheduler is applied to the learning rate to gradually decrease it over training epochs, promoting stable convergence:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min})\left(1 + \cos\left(\frac{T_{cur}}{T_{max}}\pi\right)\right) \tag{11}$$

where,

$$\eta_{\max} = \text{Initial learning rate}$$

$$\eta_{\min} = \text{Minimum learning rate}$$

$$T_{\text{cur}} = \text{Current epoch}$$

$$T_{\max} = \text{Total number of epochs}$$

3.5.3 Loss function: Joint classification and domain adaptation

The total loss $\mathcal{L}_{total}$ is a weighted sum of the classification loss and the domain adaption loss.

- Classification Loss: Cross entropy for multi-class tumor prediction.

$$\mathcal{L}_{cls} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{c=1}^{C} y_{i,c}\log p_{i,c} \tag{12}$$

  where, $y_{i,c}$ is the ground truth label and $p_{i,c}$ is the predicted probability for class c.

- Domain Adaption Loss: Binary cross entropy for domain discrimination.

$$\mathcal{L}_{domain} = -\frac{1}{B}\sum_{i=1}^{B}[d_i \log \widehat{d}_i + (1 - d_i)\log(1 - d_i)] \tag{13}$$

  where, d_i is the domain label and $\widehat{d}_i$ is the predicted domain confidence.

- Total Loss:

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \lambda_{domain} \cdot \mathcal{L}_{domain} \tag{14}$$

3.5.4 Progressive fine-tuning

A progressive fine-tuning approach is used to stabilize training and avoid devastating forgetting. In the early epochs, all backbone layers (ResNet and ViT) are frozen, leaving just the fusion, classifier, and local cross-attention modules trainable. Deeper layers of the ResNet backbone are progressively unfrozen and their learning rate is raised after a predetermined epoch E1, enabling the model to progressively adapt increasingly complicated convolutional features to the target task. The ViT backbone is then likewise unfrozen at a later epoch, E2, allowing for combined end-to-end training of the complete architecture. Before the model fine-tuning, all layers, this staged unfreezing technique, make sure that high-level task-specific features are learnt first, giving it a solid foundation and enhancing convergence and generalization overall.

3.5.5 Training and validation performance analysis

The accuracy and loss curves for training and validation over 50 epochs are shown in Fig. 6. With an initial accuracy of 82.58% and a comparatively large loss of 0.62, the training process starts. The model rapidly converges as training goes on, with accuracy approaching 100% by epoch 21 and the training loss gradually declining. Strong generalization capacity on unknown data is demonstrated by the validation accuracy, which likewise steadily increases, reaching a peak of 98.95% after only a few epochs and nearing 98%. The validation loss varies throughout epochs, but the training loss exhibits a steady decline. Particularly after epoch 20, these oscillations suggest either overfitting or mild instability. The validation accuracy, however, is continuously high, indicating

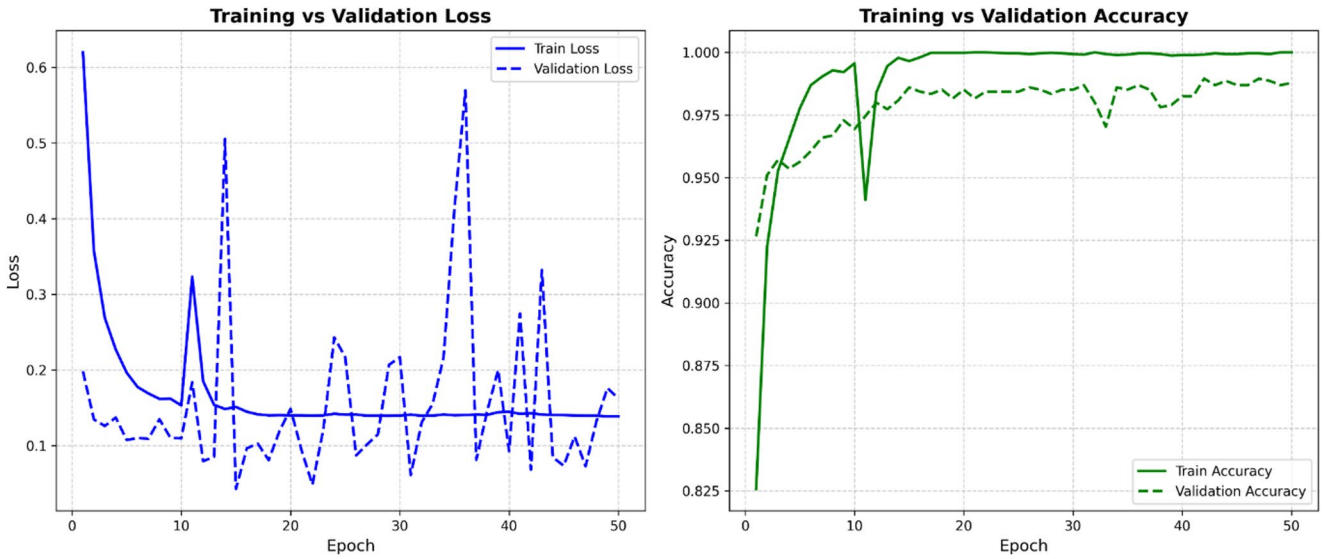


Fig. 6 Training graph of the proposed RViT-FusionNet model

that the model can still produce correct predictions in spite of the chaotic validation loss curve. Remarkably, sporadic increases in validation loss have no discernible impact on overall accuracy, underscoring the robustness of the model.

3.5.6 Checkpointing

At the end of each epoch, the model's validation accuracy is evaluated. The weights are saved whenever a new best validation accuracy is achieved:

$$If\ Acc_{val}^{(t)} > Acc_{val}^{best} : save\left(\theta^{(t)}\right) \tag{15}$$

where, $\theta^{(t)}$ are the model parameters at epoch t, and Acc_{val}^{best} is the best recorded validation accuracy.

3.6 Model testing

We trained the RViT-FusionNet architecture on the DS1 training set, saved the best model checkpoint based on validation performance, and then thoroughly tested it on other datasets to assess its generalizability and robustness. To find out how well the model operates outside of the original data distribution, specifically tested on the held-out test set of DS1, as well as three completely unseen datasets, DS2, DS3, and DS4. To provide transparency and further comprehend the proposed model's decision-making process, we used Grad-CAM for tumor localization visualization, emphasizing the areas of MRI scans that made the biggest contributions to the forecasts. The model's performance was assessed using common metrics such as accuracy, precision, recall, and F1-score, which, when taken as a whole, offer a thorough evaluation. Due to the inherent imbalance in these datasets, some tumor classes have far more samples than others—accuracy by itself can be deceptive because it may disproportionately represent the majority classes. As a result, further metrics, including precision, recall, and F1-score, were included to guarantee a fair and significant assessment, providing more information about the model's capacity to accurately detect both common and uncommon tumor forms. The ratio of accurately predicted cases to all cases is used to calculate accuracy in this context:

- Accuracy: Accuracy calculates the ratio of accurately predicted instances to total instances, which indicates how accurate the model is overall. The equation for accuracy:

$$Accuracy = \frac{\sum_{i=1}^{N} TP_i}{\sum_{i=1}^{N} TP_i + FP_i + FN_i} \tag{16}$$

Here, TP_i = true positive for class i, FP_i = false positive for class i, and FN_i = false negative for class i.

- Precision: Another word for precision is Positive Predictive Value (PPV), which measures the percentage of cases that are actually correct when classified. A high precision score indicates the lack of frequent misclassifications of other classes as the target class. For precision, the equation is:

$$Precision = \frac{TP}{TP+FP} \tag{17}$$

Here, TP = true positive and FP = false positive.

- Recall: The efficacy of the model in identifying every genuine instance of a particular class is quantified by recall, which is frequently denoted as sensitivity or true positive rate (TPR). A diminished recall signifies that numerous accurate instances are overlooked (resulting in elevated false negatives), whereas an elevated recall indicates that the model adeptly classifies the preponderance of pertinent instances. The equation for recall is:

$$Recall = \frac{TP}{TP+FN} \tag{18}$$

Here, TP = true positive and FN = false negative.

- F1-score: The F1-score represents a comprehensive metric that incorporates both false positives and false negatives into its evaluation framework. It is derived as the harmonic mean of Precision and Recall. Due to its capacity to ensure that neither accuracy nor recall predominates in the assessment, it proves to be particularly advantageous when dealing with imbalanced datasets. A high F1-score indicates that the model serves as a reliable measure for assessing classification performance, as it demonstrates proficiency in both domains.

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recal} \tag{19}$$

4 Results

4.1 Model's performance

The test results of the suggested RViT-FusionNet model for classifying brain tumors are shown and discussed in this section. The performance evaluation is conducted for a single run, using multiple datasets to assess the model's effectiveness across four categories: pituitary, meningioma, glioma, and no tumor. We used common evaluation criteria, such as accuracy, precision, recall, and F1-score, to test the model under a variety of scenarios and ensure a balanced assessment that went beyond only accuracy. The model with the highest validation accuracy during training was saved. When evaluated on the test set of DS1, comprising 1,311 MRI images, the RViT-FusionNet achieved an impressive accuracy of 99.16%. As shown in Fig. 7(a), most samples across the four categories are correctly classified, with very few misclassifications between meningioma and pituitary tumors. On DS2 (7,153 images), the model attained 99.66% accuracy. Figure 7(b) shows that almost all images were

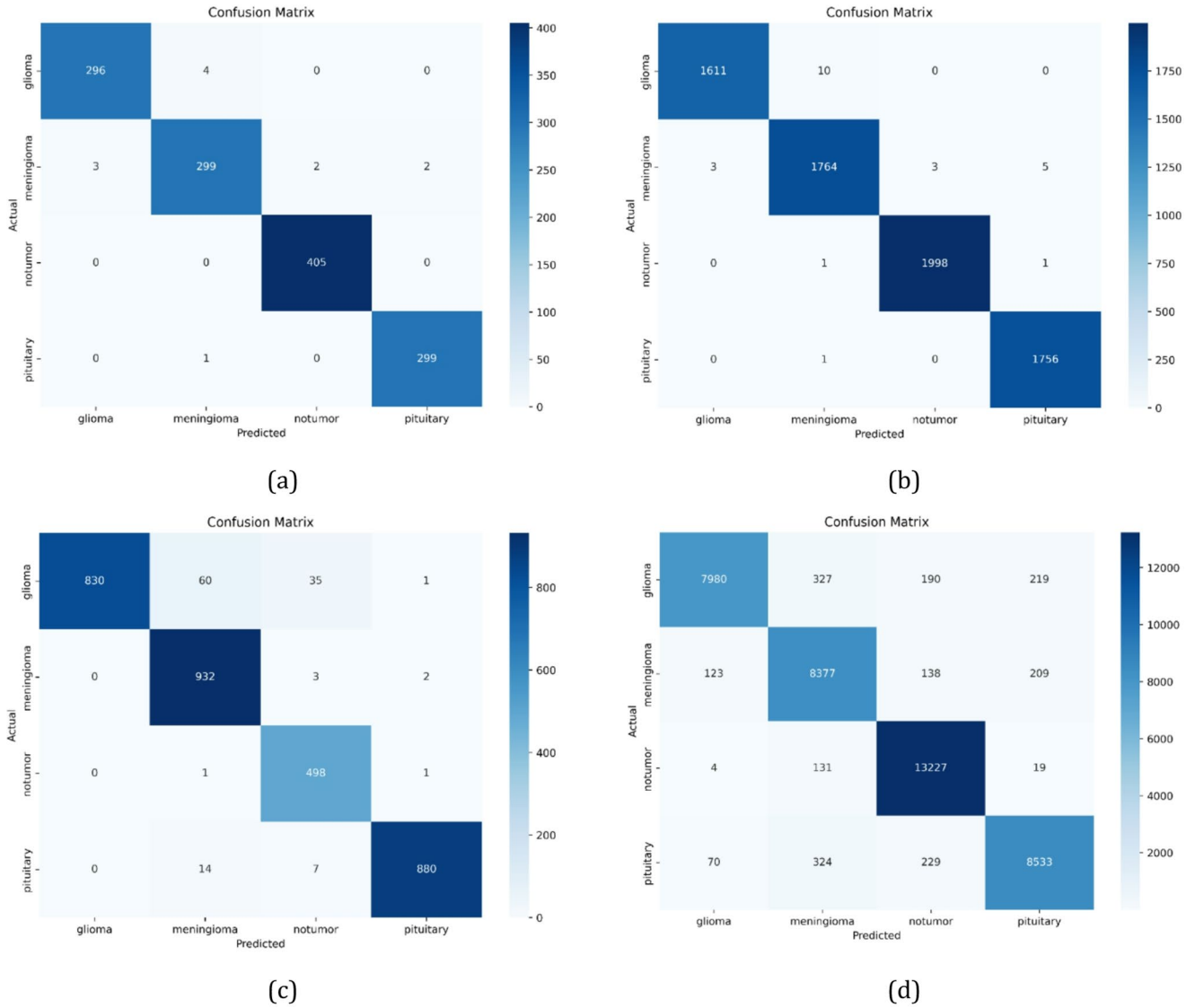


Fig. 7 Confusion matrix analysis for multiple datasets. (**a**) DS1, (**b**) DS2, (**c**) DS3, and (**d**) DS4

predicted correctly, with negligible errors across the tumor types, further confirming the robustness of the proposed method. For DS3 (3,264 images), the model achieved 96.20% accuracy. The confusion matrix in Fig. 7(c) reveals slightly higher misclassifications compared to DS1 and DS2, particularly between glioma and meningioma, as well as between pituitary and no tumor classes. Finally, on DS4 (40,100 images), the RViT-FusionNet reached 95.05% accuracy. As illustrated in Fig. 7(d), although the classification performance remains strong, the larger dataset and greater variability in images resulted in more noticeable misclassifications, particularly among glioma, meningioma, and pituitary categories.

In Table 3, the classification results show consistently strong performance in differentiating between pituitary tumors, meningioma, glioma, and no tumor cases across the four datasets (DS1, DS2, DS3, and DS4). The overall accuracy for DS1 is 99.16%, with the model achieving near-perfect precision, recall, and F1-scores for all classes. This suggests that the dataset is highly reliable, with very little misclassification. Although the measures remain strong in DS2, DS3, and DS4, which were completely unseen by the model and not used for training or fine-tuning. The performance shows a slight decline, particularly for the glioma and meningioma classes, where recall and F1-scores are comparatively lower. Nevertheless, the model sustains strong precision and recall for

Table 3 Classification report analysis for multiple datasets

Dataset	Class	Precision	Recall	F1-Score	Accuracy
DS1	Glioma	1.00	0.99	0.99	99.08%
	Meningioma	0.98	0.98	0.98	
	No tumor	0.99	1.00	1.00	
	Pituitary	0.99	1.00	1.00	
DS2	Glioma	1.00	0.99	1.00	99.66%
	Meningioma	0.99	0.99	0.99	
	No tumor	1.00	1.00	1.00	
	Pituitary	1.00	1.00	1.00	
DS3	Glioma	1.00	0.90	0.95	96.20%
	Meningioma	0.93	0.99	0.96	
	No tumor	0.92	1.00	0.95	
	Pituitary	1.00	0.98	0.99	
DS4	Glioma	0.98	0.92	0.94	95.05%
	Meningioma	0.91	0.95	0.93	
	No tumor	0.96	0.99	0.97	
	Pituitary	0.95	0.93	0.94	

the no tumor and pituitary classes, while overall accuracy remains above 95% for these datasets. These results highlight the model's robustness and generalizability across multiple datasets, with only negligible performance differences.

The visual results in Fig. 8 illustrate the accurate predictions of the proposed RViT-FusionNet model on brain MRI scans. Here, the model successfully identifies glioma, meningioma, pituitary tumors, and no tumor cases with high confidence, as reflected by the near-100% predicted probabilities for the true class. This demonstrates the model's ability to distinguish between tumor types and healthy cases.

4.2 Ablation and statistical significance analysis

We carried out an ablation analysis to verify the distinct contributions of the RViT-FusionNet components. The statistical significance analysis of the DS1 dataset, which quantifies the contributions of the Domain Discriminator and Local Cross-Attention (LCA) modules, is shown in Table 4. The proposed RViT-FusionNet achieved the highest mean accuracy with low standard deviation across 10 independent runs ($n=10$), demonstrating superior stability. Welch's t-tests confirm that removing the LCA module results in a statistically significant performance drop ($t=3.4049$, $p=0.00398$), validating its importance for cross-modal feature interaction. The most critical degradation occurred upon removing the Domain Discriminator ($t=5.2386$, $p=0.0017$), indicating that the stochastic regularizer-based discriminator is the primary driver of feature generalization and robustness. Furthermore, the removal of both modules yielded a significant decline ($t=2.4162$, $p=0.03301$), confirming that the LCA and domain discriminator modules provide essential, complementary performance gains.

For further analysis, we generated box plots to observe the accuracy variations across multiple runs of the different RViT-FusionNet variants on the DS1 dataset, as shown in Fig. 9. The RViT-FusionNet model achieves the highest median accuracy with the lowest variance, demonstrating both superior performance and strong stability. Removing the LCA module leads to a noticeable drop in median accuracy, highlighting its importance in capturing fine-grained contextual features. The w/o Domain Disc variant produces the lowest median accuracy, indicating that domain discrimination is critical for maintaining baseline performance. When both LCA and the domain discriminator are removed, performance partially recovers relative to the single-component removal case but remains inferior and less stable than the RViT-FusionNet model, confirming the complementary contributions of both modules. However, when we tested the variant without both modules on the DS4 dataset, which is the largest dataset, it did not perform well and failed to accurately classify the brain tumor MRI images.

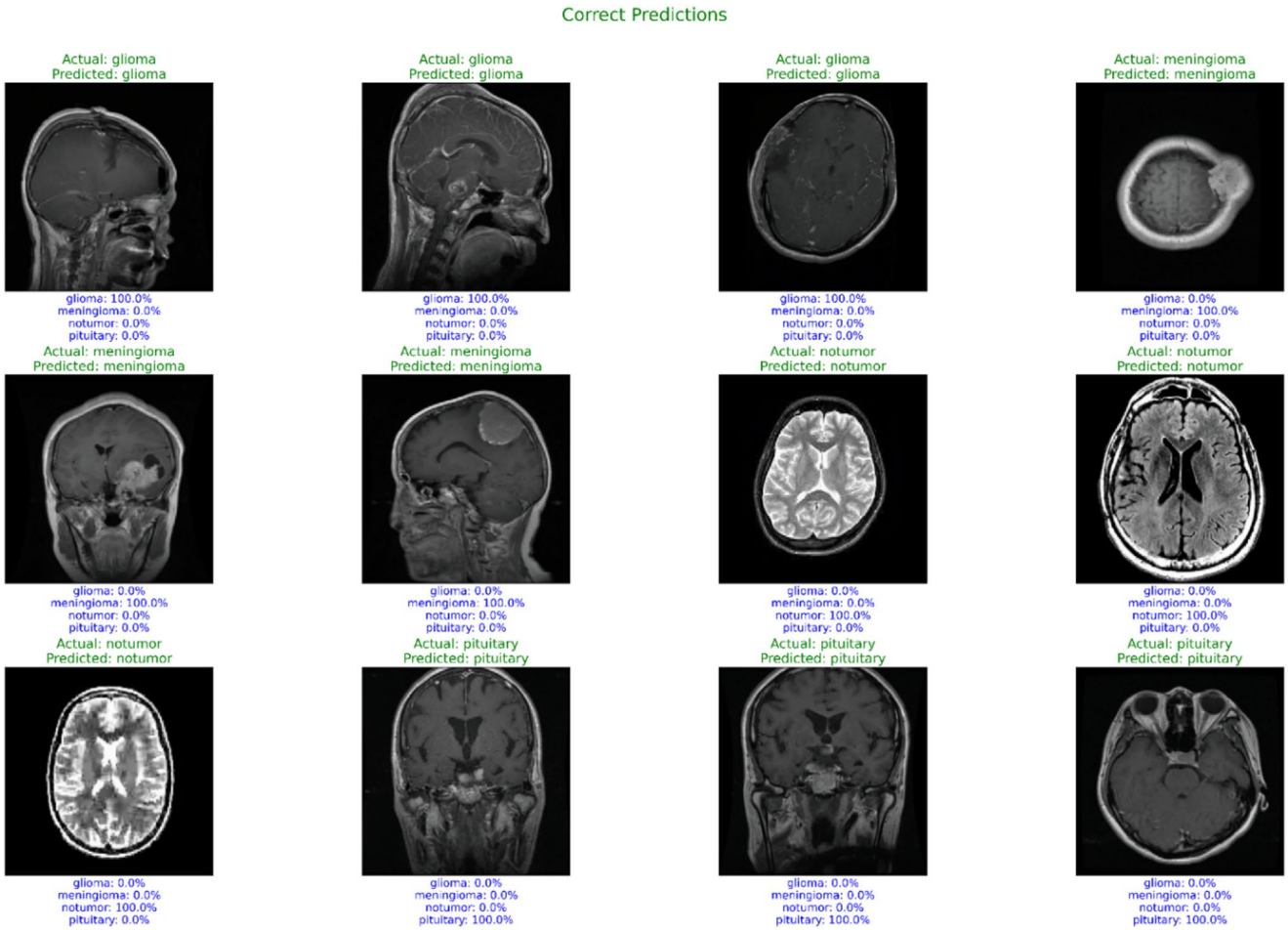


Fig. 8 Visualization of the RViT-FusionNet model's accurate predictions on brain MRI scans

Table 4 Statistical significance analysis of RViT-FusionNet components compared to ablated variants

Model Variant	Result (Mean ± Std)	t-statistic	*p*-value	Significant ($p<0.05$)?
RViT-FusionNet (Proposed Method)	0.9908 ± 0.0016	---	---	---
w/o Local Cross-Attention (LCA) Module	0.9875 ± 0.0026	3.4049	0.00398	Yes
w/o Domain Discriminator Module	0.9847 ± 0.0033	5.2386	0.00017	Yes
w/o Both Module	0.9875 ± 0.0040	2.4162	0.03301	Yes

5 Comparative study

In recent years, numerous hybrid deep learning models have been proposed for brain tumor classification, demonstrating impressive accuracy and robustness on MRI datasets. A comparative analysis for brain tumor classification is presented in Table 5. This table summarizes key attributes of each model, including the architecture used, dataset, total number of images, and reported classification accuracy. This study made use of four distinct publicly accessible datasets. The results of several models based on DS1 and DS3, which contain 7,023 and 3,264 images, respectively, are compared with the suggested architecture, RViT-FusionNet. Priya & Vasudevan [41] presented a hybrid AlexNet–GRU model for multi-class brain tumor detection and classification using the DS1 dataset. This approach utilizes AlexNet for deep spatial feature extraction, followed by a GRU layer to capture sequential dependencies across slices. This combination achieved with

 https://doi.org/10.1007/s00521-026-12290-x

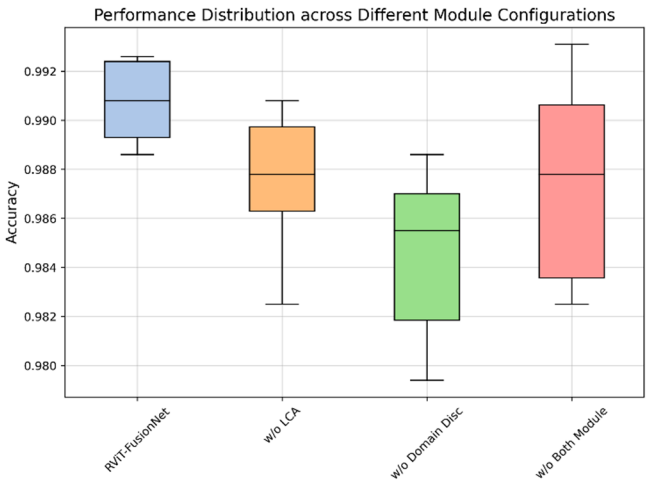


Fig. 9 Statistical analysis of the RViT-FusionNet model using box plots illustrating the accuracy distribution over 10 independent runs

Table 5 Comparing the brain tumor classification performance of the proposed method with existing methods

Dataset	Total Images	Model Name	Evaluation Metrics			
			Accuracy	Precision	Recall	F1-Score
DS1	7023	AlexNet-GRU [41]	97%	97.63%	96.78%	97.25%
		H-QNN approach [42]	95.58%	---	---	95.58%
		DenseNet121 model [43]	96.90%	97.20%	97.15%	97.10%
		CNN+LSTM model [44]	97.86%	---	---	---
		ResNet101-CWAM [23]	99.83%	99.01%	99.21%	99.27%
		RViT-FusionNet	**99.08% ± 0.16%**	**99.08% ± 0.10%**	**99.00% ± 0.24%**	**99.07% ±0.16%**
DS3	**3264**	2D CNN model [45]	96.47%	---	95%	---
		MobDense model [46]	96.02%	96.3%	95.7%	95.9%
		DBTNet [47]	97.90%	---	---	---
		RViT-FusionNet	**96.20% ± 0.25%**	**96.20% ± 0.24%**	**96.21% ± 0.23%**	**96.23% ±0.20%**
Other's Dataset						
BrTMHD-2023	1166	ViT+GRU [48]	98.97%	---	---	97%
BraTS-2018	1425	TECNN [49]	96.75%	---	---	
Brain MRI dataset	3064	Hybrid BTA [50]	98.62%	98.88%	97.53%	98.19%
DS2	**7153**	**RViT-FusionNet**	**99.56% ± 0.17%**	**99.95% ± 0.03%**	**99.95% ± 0.04%**	**99.98% ±0.02%**
DS4	**40,100**	**RViT-FusionNet**	**94.76% ± 0.35%**	**94.83% ± 0.17%**	**94.85% ± 0.18%**	**94.89% ±0.12%**

an accuracy of 97%, precision of 97.63%, recall of 96.78%, and an F1-score of 97.25%, demonstrating the benefits of coupling convolutional and recurrent networks for extracting both spatial and temporal patterns. However, training was limited to a single public dataset, raising potential issues with overfitting and generalizability. Moreover, while the model offers strong classification performance, its reliance on GRU increases computational complexity and may complicate deployment in real-time or resource-constrained settings.

Ranga et al. [42] used ZZFeatureMap and RealAmplitudes Ansatz to combine convolutional layers with quantum neural networks to propose a hybrid quantum-classical neural network (H-QNNs) for brain tumor classification. By addressing data imbalance through augmentation and weighted loss, this method demonstrated the promise of quantum machine learning in medical imaging and achieved 95.58% accuracy, with an F1-score of 95.58% on the DS1 dataset. Rasheed et al. [43] investigated the use of DenseNet121 with transfer learning for brain tumor identification in MRI images. The model outperformed traditional diagnostic techniques and demonstrated the effectiveness of automated deep learning approaches in medical imaging by applying preprocessing and classifying tumors into benign, glioma, meningioma, and pituitary categories. This model achieved 96.90% accuracy, with a precision of 97.20%, recall of 97.15%, and an

F1-score of 97.10% on the DS1 dataset. Abbas et al. [44] developed a hybrid model combining a Convolutional Neural Network (CNN) with a Long Short-Term Memory (LSTM) network for brain tumor classification. Using the DS1 dataset comprising 7,023 MRI images across four tumor classes, the CNN–LSTM model achieved an accuracy of 97.86%. Using the DS1 dataset, Srinivasan et al. [23] achieved an accuracy of 99.83% with their ResNet101-CWAM model, along with a precision of 99.01%, a recall of 99.21%, and an F1-score of 99.27%. However, the integration of CWAM modules increased computational complexity, model size, and memory requirements, making deployment more resource-intensive. The suggested approach, RViT-FusionNet, is a unique hybrid deep learning model that improves the classification accuracy of brain tumors by combining Convolutional Neural Networks (CNNs) with Vision Transformers (ViTs) via a local cross-attention mechanism. ViTs are used for global contextual comprehension, and CNNs for local feature extraction in this fusion, which produces better classification capabilities. 99.08% ± 0.16% accuracy was attained by RViT-FusionNet when tested on the DS1 dataset. To be more precise, the model achieved strong overall performance, with a precision of 99.08% ± 0.10%, a recall of 99.00% ± 0.24%, and an F1-score of 99.07% ± 0.16%, indicating highly reliable classification across all tumor categories. These findings demonstrate the durability and dependability of RViT-FusionNet, markedly outperforming traditional methods and demonstrating its potential for use in automated brain tumor diagnosis in clinical settings.

This study made use of the DS3 dataset, which included 3,264 MRI pictures. Saeedi et al. [45] proposed that 2D CNN outperformed several conventional machine learning techniques while retaining low complexity and quick execution appropriate for clinical use. It was able to classify gliomas, meningiomas, pituitary tumors, and healthy brains with an average recall of 95% and a training accuracy of 96.47%. The MobDense model was proposed by Meher et al. [46] for the classification of brain tumors using the DS3 dataset. This model achieved a rather low accuracy of 96.02%, with a precision of 96.3%, recall of 95.7%, and an F1-score of 95.9%. Reddy et al. [47] introduced DBTNet, a dual-stream neural network designed for brain tumor detection from MRI images. Their model achieved 97.90% accuracy on the DS3 dataset. But our proposed architecture RViT-FusionNet, achieved 96.20% ± 0.25% accuracy with a precision of 96.20% ± 0.24%, a recall of 96.21% ± 0.23% and an F1-score of 96.23% ± 0.20% on the DS3 dataset. The comparison between DBTNet and RViT-FusionNet shows that DBTNet is much more complex than RViT-FusionNet.

Ahmed et al. [48] employed the BrTMHD-2023 dataset containing 1,166 MRI images and achieved an accuracy of 98.97% and an F1-score of 97% using a ViT + GRU-based framework. Similarly, Aloraini et al. [49] applied a TECNN model on the BraTS-2018 dataset with 1,425 images, obtaining an accuracy of 96.75%. Prakram et al. [50] introduced a Hybrid BTA model on a brain MRI dataset of 3,064 images and reported an accuracy of 98.62%, a precision of 98.88%, a recall of 97.53% and an F1-score of 98.19%. In comparison, our proposed RViT-FusionNet demonstrated superior performance, achieving 99.56% ± 0.17% accuracy with a precision of 99.95% ± 0.03%, a recall of 99.95% ± 0.04% and an F1-score of 99.98% ± 0.02% on the DS2 dataset with 7,153 images. The suggested RViT-FusionNet model achieved an accuracy of 94.76% ± 0.35% with a precision of 94.83% ± 0.17%, a recall of 94.85% ± 0.18% and an F1-score of 94.89% ± 0.12% on the larger DS4 dataset comprising 40,100 images, thereby outperforming existing methods across diverse datasets.

6 Discussion

In this research, RViT-FusionNet is a robust and explainable deep learning framework for multi-class brain tumor classification using MRI scans. The model leverages the strengths of transformer-based architectures and multi-modal feature fusion, enabling it to effectively capture both global and local contextual information from complex medical images. Its transformer-based encoder processes images via patch embedding, which makes the architecture fundamentally scalable to higher-resolution inputs by adjusting the patch size

or sequence length. However, to maintain computational efficiency and manageable GPU memory usage during testing with very high-resolution volumes (e.g., above 256 × 256), resizing is typically applied. This resizing balances preserving fine anatomical detail with hardware constraints, ensuring stable training and effective accuracy.

Extensive experimental results across multiple datasets demonstrate the high accuracy, precision, recall, and F1-scores achieved by RViT-FusionNet, confirming its reliability and generalizability in distinguishing between glioma, meningioma, pituitary tumors, and non-tumor cases. To further enhance the interpretability of the proposed model, Grad-CAM-based tumor localization was utilized, as illustrated in Fig. 10. The Grad-CAM visualizations provide valuable insights into the decision-making process of RViT-FusionNet by highlighting the specific regions in the MRI scans that most influenced the model's predictions. In the presented examples, the heatmaps are well-aligned with the actual tumor locations, demonstrating that the model consistently focuses on clinically relevant areas when making its predictions. This level of transparency is crucial for building trust in AI-assisted diagnostic tools and can aid clinicians in understanding and validating automated predictions. Overall, the combination of high classification performance and strong localization capability underscores the potential of RViT-FusionNet as a reliable and interpretable tool for brain tumor diagnosis.

Therefore, the suggested method was evaluated on multiple datasets containing varying numbers of images. The model effectively classified four distinct types of brain tumors with high accuracy. However, some misclassifications were observed, primarily due to class imbalance within the datasets and variations in image quality.

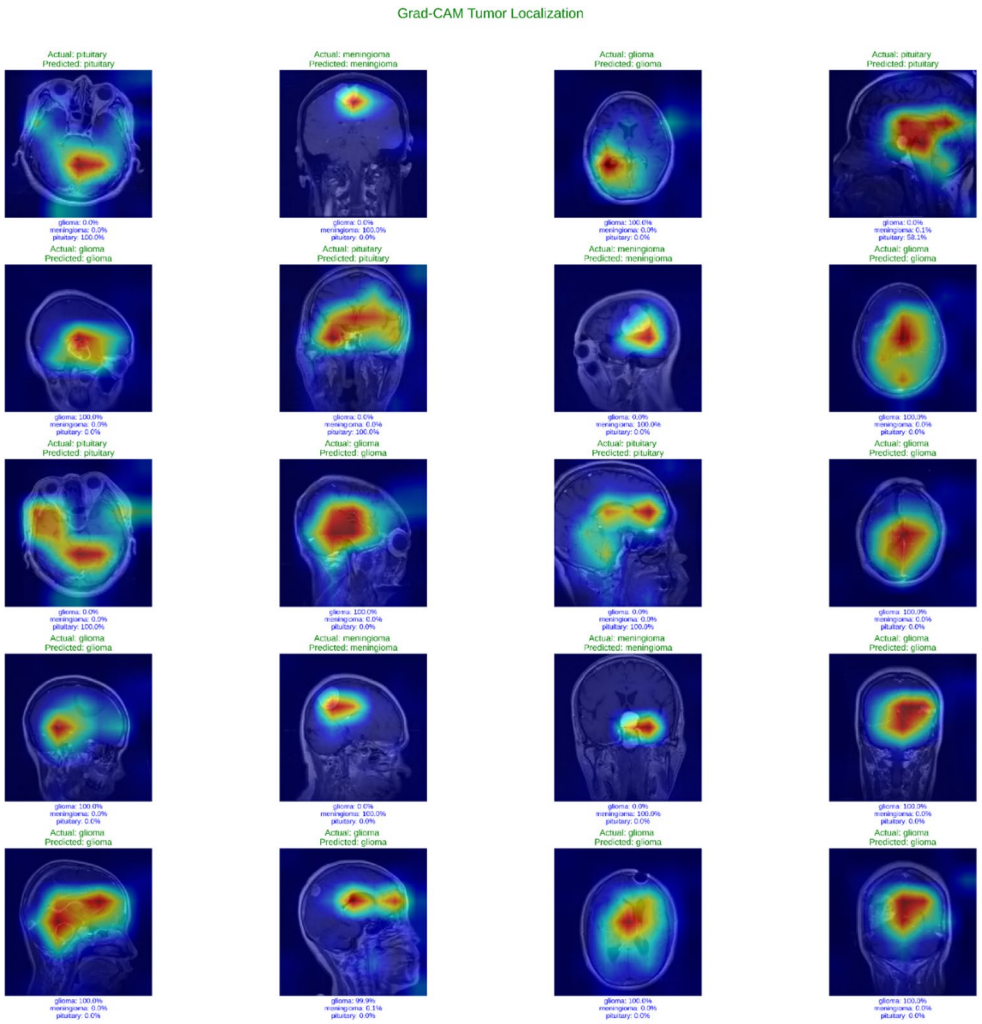


Fig. 10 Grad-CAM Localization of Brain Tumor Classification (best seen in color)

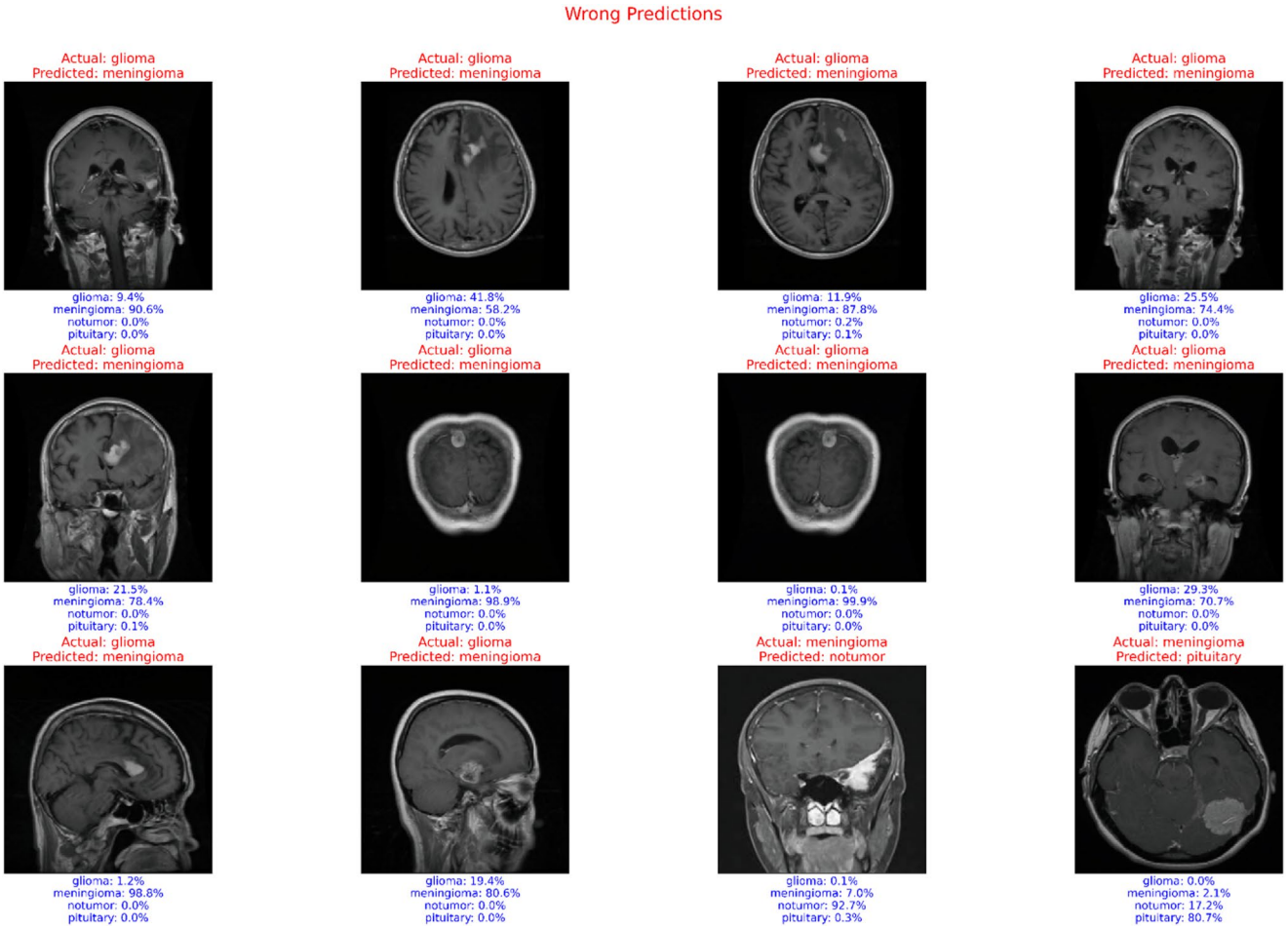


Fig. 11 Samples of incorrect predictions where the model misclassifies, the confidence scores help in understanding the misclassifications, such as glioma being mislabeled as pituitary or meningioma (best seen in color)

These factors affected the model's ability to distinguish between certain tumor types in edge cases. Sample images of such misclassifications are illustrated in Fig. 11.

7 Conclusion

RViT-FusionNet is an advanced hybrid deep learning architecture for classifying brain tumors, including gliomas, meningiomas, pituitary, and healthy tissue, by combining the local texture extraction of ResNet-50 with the global context of a Vision Transformer and combining them through a Local Cross-Attention module. An adversarially trained patch-based domain discriminator guarantees domain-invariant features across a variety of MRI protocols. The model was trained on DS1 and evaluated on several independent datasets, including the DS2, DS3, and DS4 datasets, and achieved high accuracy, precision, recall, and F1-scores. Grad-CAM visualizations verified that the network concentrates on clinically relevant tumor regions. In practice, RViT-FusionNet can be used as a second-opinion tool in radiology workstations to expedite diagnosis and decrease variability. Precise subtype classification and tumor boundary localization can guide pre-surgical planning, longitudinal treatment response monitoring can be enabled in neuro-oncology follow-ups, and expert-level diagnostic support can be extended to remote or resource-constrained settings through cloud or embedded imaging solutions. As part of future work, we plan to refine fusion and class-aware loss

functions, integrate multi-modal sequences (T1, T2, FLAIR, and contrast-enhanced), and extend RViT-FusionNet to full 3D volumetric MRI by incorporating three-dimensional convolutions and attention mechanisms. Furthermore, we intend to broaden our quantitative evaluation by including a more comprehensive set of metrics such as PSNR and SSIM for structural fidelity, along with information-theoretic measures like Mutual Information, Qab/f, and entropy to enable a more rigorous and multi-faceted assessment of fusion quality. These efforts aim to further reduce misclassifications and enhance the clinical applicability of the framework.

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Data availability The datasets are analyzed during this study are available in the [DS1 datasets], [DS2 datasets], [DS3 datasets] and [DS4 datasets] repositories.
- DS1 dataset: https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset.
- DS2 dataset: https://www.kaggle.com/datasets/tombackert/brain-tumor-mri-data.
- DS3 dataset: https://www.kaggle.com/datasets/sartajbhuvaji/brain-tumor-classification-mri/data.
- DS4 dataset: https://doi.org/10.34740/kaggle/dsv/10329066.

Declarations

Competing interests The authors declare no competing interests.

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